

MAJOR PROJECT SYNOPSIS & PUBLICATION PROPOSAL

ShareBite

A Smart Food Sharing & Real-Time NGO Distribution Platform

PROJECT SYNOPSIS

Project Title: ShareBite: Surplus Food Recovery & NGO Redistribution Platform

Course / Degree: Master of Computer Applications (MCA)

Project Mentor: Ms. Perna Awasthi (Assistant Professor / Project Guide)

Project Team:

Student Name	University Roll No
Varun Kumar	2500910140036
Neha Khatoon	2500910140026

Core Tech Stack: React 19, Node.js, Express 5, PostgreSQL, Leaflet Maps, OpenStreetMap

Academic Session: 2026

Department of Master of Computer Applications

Final Year MCA Project & Research Initiative | ShareBite Platform

1. PROJECT ABSTRACT (SIMPLE SUMMARY)

Every day, large quantities of fresh, good-quality food are thrown away by restaurants, banquet halls, wedding caterers, and bakeries. At the same time, millions of underprivileged people, homeless individuals, and children go to sleep hungry. The main reason this happens is not a shortage of food, but a **lack of communication and quick connection** between the people who have extra food and the NGOs or charities that can distribute it.

ShareBite is an easy-to-use web platform that solves this problem in real time. Whenever a restaurant or caterer has extra food, they can post it on ShareBite in less than 30 seconds using automatic GPS location. Nearby verified NGOs and volunteers instantly see the food on an interactive map. An NGO volunteer can claim the food batch and receive a secure **6-digit OTP code**. When the volunteer reaches the restaurant, the donor checks this OTP before handing over the food. To ensure that nobody ever receives spoiled food, an automatic safety timer checks all listings every minute and immediately removes any food that has passed its safe expiry time.

Core Benefit: ShareBite connects food donors with nearby NGOs in **under 12 minutes**, preventing food waste, stopping food poisoning, and feeding hungry people.

2. PROBLEM STATEMENT (WHY THIS PROJECT IS NEEDED)

Currently, donating surplus food is difficult because of several real-world issues:

- **Food Spoils Quickly:** Cooked food stays fresh for only 3 to 5 hours. In traditional systems, people make phone calls or send WhatsApp messages, which takes too long, and the food often goes bad before anyone collects it.
- **Donors Don't Know Nearby NGOs:** A restaurant manager rarely knows which nearby charity or soup kitchen has free volunteers or hungry people to feed right now.
- **Safety & Trust Issues:** Without a verification system, anyone could take the food for the wrong reasons, and donors worry about liability or food safety.
- **No Automatic Expiry Control:** Ordinary websites do not automatically remove old food, which can cause food poisoning if expired food is consumed.

3. OBJECTIVES OF THE PROJECT

The key goals of ShareBite are:

1. **Quick Food Posting (Under 30 Seconds):** Allow restaurants and donors to easily post extra food with quantity (servings/kg/boxes), food type (Veg/Non-Veg), and safe expiry time.
2. **Live Map & Nearby NGO Search:** Show all available food batches and nearby NGOs on an interactive map using real-time GPS within an adjustable radius of 2 to 50 km.
3. **Safe Pickup Using 6-Digit OTP:** Protect the food from theft or wrong delivery by generating a unique 6-digit OTP when a food batch is claimed. The donor enters this OTP before handing over the food.
4. **Automatic Food Safety Timer:** Run an automatic check every 60 seconds. If a food item reaches its safe-by expiry time, the system automatically marks it as expired so no one can claim spoiled food.
5. **Turn-by-Turn Driving Directions:** Show the volunteer the best driving route, distance (km), and estimated travel time (minutes) to reach the donor's location quickly.
6. **Measurable Impact & Certificates:** Automatically track the total number of meals saved and kilograms of food rescued, and generate a printable Certificate of Impact for donors.

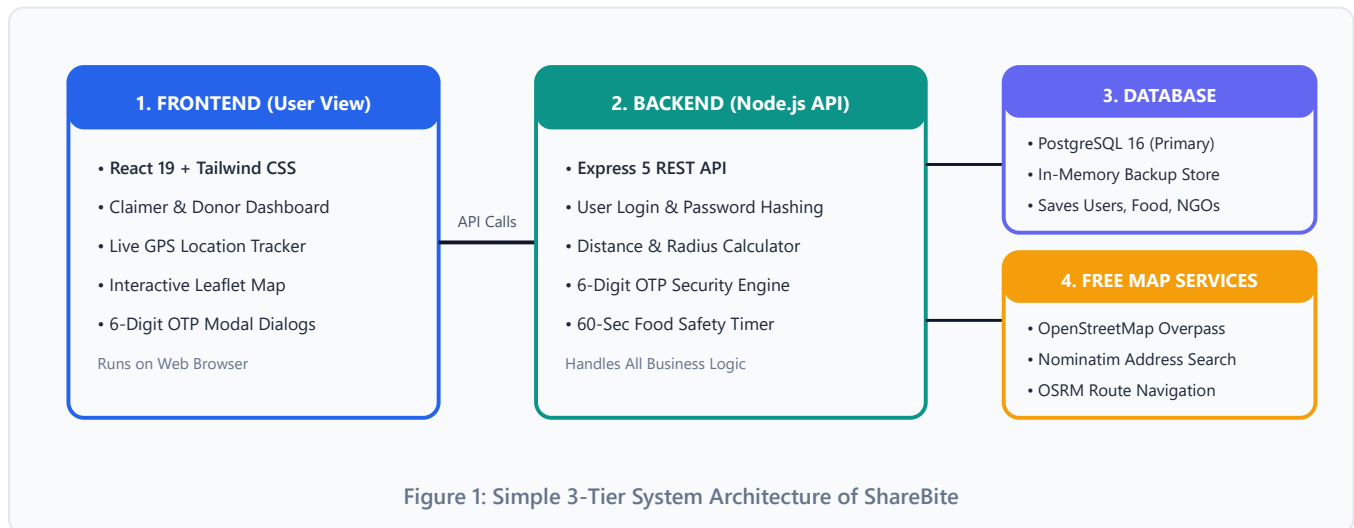
4. COMPARISON WITH EXISTING SYSTEMS

Here is how ShareBite compares with other food apps and traditional methods:

Feature	Too Good To Go	Robin Hood Army	Olio	ShareBite (Our Project)
Purpose	Sells cheap food to customers	Manual volunteer drives	Neighbour food sharing	Free Real-Time Food Rescue for NGOs
Live GPS Map	Commercial map	No map (WhatsApp groups)	Zip code list	Interactive Leaflet Map (2-50 km radius)
Pickup Security	Store bill screen	Trust / None	Direct chat	Secure 6-Digit Two-Party OTP
Auto-Expiry Safety	Store closing time	Manual check only	User report	Automatic 60-Second Safety Cron Job
Database Setup	Cloud only	Excel / WhatsApp	Cloud only	PostgreSQL + Zero-Config In-Memory Mode
Impact Certificate	Annual email	None	Simple counter	Instant Printable Official Certificate

5. SYSTEM ARCHITECTURE (HOW IT WORKS)

ShareBite uses a clean and modern 3-tier architecture that is fast, reliable, and easy to maintain:



5.1 Main Modules of ShareBite

- **Food Posting Module:** Donors enter the food title, whether it is Veg or Non-Veg, quantity, and when it was prepared. The app uses GPS to fill the address automatically.
- **Nearby Map & Search Module:** Shows orange markers for food batches, blue markers for NGOs, and green markers for restaurants. Users can filter by distance (2 km to 50 km).
- **OTP Pickup Verification Module:** Generates a random 6-digit code for the NGO. The donor enters this code to confirm that the real volunteer has collected the food.
- **Automated Food Safety Module:** Checks all food listings every minute. If any food is past its safe consumption time, it is removed from the map automatically.
- **Admin Dashboard Module:** Allows college or municipal supervisors to check total meals saved and verify new NGOs and donors with one click.

6. KEY FORMULAS USED IN THE PROJECT

6.1 Real Distance Calculation (Haversine Formula)

To find the exact distance between the user's phone GPS and nearby restaurants or NGOs across the curve of the Earth, ShareBite uses the **Haversine Formula**:

$$\text{Distance (d)} = 2 \times R \times \arcsin(\sqrt{\sin^2(\Delta\text{lat} / 2) + \cos(\text{lat1}) \times \cos(\text{lat2}) \times \sin^2(\Delta\text{lon} / 2)})$$

where Earth Radius R = 6,371 km

This allows the app to sort food listings by distance instantly in less than 20 milliseconds.

6.2 Food Rescued and Carbon Savings Formula

When food rots in landfills, it creates harmful greenhouse gases. ShareBite calculates total meals saved and carbon prevented:

$$\begin{aligned} \text{Total Meals Saved} &= \text{Sum of all servings distributed} \\ \text{Total Rescued Weight (kg)} &= (\text{Servings} \times 0.40 \text{ kg}) + (\text{Boxes} \times 2.50 \text{ kg}) + \text{direct kg} \\ \text{Carbon Prevented (kg CO}_2\text{e)} &= \text{Total Rescued Weight (kg)} \times 2.50 \text{ kg CO}_2\text{e / kg} \end{aligned}$$

7. PROJECT IMPLEMENTATION PLAN & STATUS

Phase	Task Details	Technologies Used	Status
Phase 1	Requirement Analysis, IEEE SRS & Design Diagrams	Software Engineering Standards	Completed
Phase 2	Database Design with PostgreSQL and In-Memory Fallback	PostgreSQL 16, Node.js	Completed
Phase 3	Backend REST API with Login, OTP, and 60-Sec Safety Timer	Express 5, node-cron, bcrypt	Completed
Phase 4	Live Map Integration with Leaflet and OpenStreetMap	Leaflet, OSM Overpass, OSRM	Completed
Phase 5	Interactive React 19 Frontend with Dark/Light Mode	React 19, Tailwind CSS v4	Completed
Phase 6	Testing, Documentation & Research Paper Writing	Chrome Headless, Vitest	Ready for Publication

8. EXPECTED OUTCOMES & SOCIAL VALUE

What ShareBite Delivers:

- **A Complete Working Web Application:** Runs smoothly on laptops, tablets, and mobile phones with zero complicated setup.
- **Feeds Hungry People:** Moves fresh, safe food from banquet halls and restaurants straight to hungry children and poor families.
- **Stops Spoilage with Timers:** Ensures no expired food is ever distributed.
- **Digital Certificates for Donors:** Provides restaurants with official proof of their food donations and environmental impact.

9. CONCLUSION & FUTURE SCOPE

ShareBite shows that modern web technologies and open-source map services can make food recovery fast, secure, and transparent. It solves the real-world communication delay between donors and charities. In the future, we plan to add:

- **AI Food Waste Prediction:** Using machine learning to help restaurants predict how much extra food they might have based on past weekend sales and weather.
- **Smart Temperature Sensors:** Using small IoT sensors inside delivery boxes to make sure food stays cool and fresh during delivery.
- **Multi-Stop Route Optimization:** Helping an NGO van collect food from 3 or 4 restaurants in one efficient trip.

10. REFERENCES

- [1] United Nations Environment Programme (UNEP), "Food Waste Index Report 2024," Nairobi, Kenya, 2024.
- [2] Food Safety and Standards Authority of India (FSSAI), "Recovery & Distribution of Surplus Food Regulations," Ministry of Health, 2019.
- [3] IEEE Std 830-1998, "IEEE Recommended Practice for Software Requirements Specifications," IEEE Computer Society.
- [4] OpenStreetMap Contributors, "Overpass API and Nominatim Documentation," 2026.
- [5] RFC 7519, "JSON Web Token (JWT) Standard," IETF, 2015.